Yucheng Pan

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Education

Ph.D. in Computer Science — Aug. 2025 – Expected May 2030
University of Maryland, College Park — College Park, MD
GPA: 3.925/4.000 | Advisors: Kaiqing Zhang and Yaodong Yu

B.Sc. in Mathematics and Physics — Sep. 2021 – Jul. 2025
Tsinghua University — Beijing, China
GPA: 3.87/4.00

Research Interests

LLM Agents and Multi-Agent Systems; Agent Evaluation and Generalization; Efficient and Long-Horizon Reasoning

Research Experience

Graduate Research Assistant — Aug. 2025 – Present
University of Maryland, College Park — College Park, MD

Rolewise Evaluation in LLM Steganography

- Led the development of a rolewise evaluation framework that separately measures encoder execution, decoder recovery, and passive-monitor detectability.
- Designed and validated controlled experiments spanning six large language models (LLMs) and eight encoding schemes; built rule-based audits to attribute outcomes to encoder, decoder, or monitor behavior.
- Showed that aggregate end-to-end recovery can mask encoding errors and decoder-specific effects, revealing distinct role-specific strengths and failure modes; led the analysis, figure development, and writing of a first-author manuscript.

Latent-State Interfaces: Alignment and Multi-Agent Communication *(ongoing)*

- Testing whether alignment toward the input-embedding space is necessary for latent reasoning, a performance stabilizer, or largely irrelevant once representation scale is controlled.
- Evaluating whether latent communication reliably transfers information that complements text and improves multi-agent reasoning or coordination.

Remote Research Intern — May 2024 – Jul. 2025
Princeton University — Remote
Mentors: Mengdi Wang (Princeton) and Huazheng Wang (Oregon State)

Deception and constraints in multi-agent systems

- Designed multi-agent experiments about deception in public goods game and constraint mechanisms based on social norms and payoff allocations; analyzed their effects on deceptive LLM-agent behavior.

Data augmentation for LLM-agent generalization

- Explored redesigning agent environments as data augmentation to improve generalization through method development and preliminary experiments.

Manuscript in Preparation

Yucheng Pan, Om Tailor, Solaleh Mohammadi, Supriya Manna, and Kaiqing Zhang.
"Beyond End-to-End Success: Rolewise Evaluation in LLM Steganography."

Technical Skills

Programming: Python, C/C++, R
Machine Learning and LLMs: PyTorch, Hugging Face Transformers, vLLM
Systems and Research Tools: Linux, Slurm, Git, LaTeX

Fellowships and Honors

Dean's Fellowship, University of Maryland — 2025–2026
Academic Excellence Scholarship, Tsinghua University — 2024
Progress in Academic Performance Scholarship, Tsinghua University — 2022
Outstanding Innovation Scholarship, Philip K. H. Wong Foundation — 2022